

Empirical Mode Decomposition of EEG Signals for Brain Computer Interface

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Abstract— Motor imagery (MI) based brain-computer interface (BCI) systems show potential applications in neural rehabilitation. In MI-BCI systems, the brain signals from movement imagination, without actual movement of limbs, can be acquired, processed and characterized to translate into actionable signals that can be used to activate external devices. However, success of such MI-BCI systems, depends on the reliable processing of the noisy, non-linear, and non-stationary brain activity signals for extraction of characteristic features for effective classification of MI activity and translation into corresponding actions. In this work, a signal processing technique, namely, empirical mode decomposition (EMD), has been proposed for processing EEG signals acquired from volunteer subjects for characterizing MI activities and activity identification.

Keywords—Brain-computer interface (BCI); electroencephalogram (EEG); empirical mode decomposition (EMD); motor imagery (MI).

I. INTRODUCTION

Brain computer interface (BCI) can be classified into two types, invasive: where electrodes are embedded into the brain surgically and non-invasive: where the measurement sensors are placed outside the head, on the scalp for instance. Most non-invasive BCIs are based on electroencephalography (EEG) [1]. Electroencephalography is a medical imaging technique that reads the scalp electrical activity generated by brain structures [2]. The EEG allows the observation of gross electrical fields of the brain, and reflects changes in neural mass activity associated with various mental processes. Some mental processes result in distinguishable EEGs, a person that can produce such mental processes at will, has the potential to use them for communication. The feasibility of this communication depends on the extent to which the EEGs associated with these mental processes can be reliably recognized automatically. Muscle activity, blinks, eye movement, and environmental noise create contamination in EEG signals [3]. Potentials from non-brain sources and channel noise are also present in EEG recording [4].

Different frequency bands of the EEG-signals correspond to different states of the human brain, hence a frequency analysis of the signals is considered very important. It gives an indication of what state the user is in. Furthermore, if

abnormalities are recorded, this can be an indication of a certain pathology, such as epilepsy [5]. Generally, one distinguishes five or six different frequency bands, also known as brain rhythms. Up to 4 Hz the rhythm is called δ -rhythm. These low frequencies have a tendency to have moderately high amplitudes up to 100 μ V. The θ -rhythm ranges from 4 to 7 Hz, α -rhythm ranges from 8-15 Hz, μ -rhythm ranges from 7.5-12.5 Hz and β -rhythm ranges from 12-30 Hz. The automatic and correct identification of the signals in these frequency bands can be used for motor imagery (MI) based BCI. In motor imagery (MI), the imagination of motor movement without actually moving any limbs, produces EEG signals from corresponding brain activities. These signals, if properly processed and identified, can be used to actuate external devices without any limb movement. The success of MI-BCI systems depend on reliable signal processing techniques suitable for brain activity related EEG signals that are characterized as noisy, nonlinear, non-stationary and with low signal-to-noise ratio.

II. EMPIRICAL MODE DECOMPOSITION (EMD)

EMD is a nonparametric and self-adaptive method which decomposes any signal to a get finite number of functions called intrinsic mode functions (IMFs) [6]. EMD does not require any predefined basis function to represent the signal, which is the advantage of EMD over Fourier and wavelet analysis. This algorithm has advantages in biomedical applications. In [7] EMD has been combined with chaos analysis to recognize the chaotic properties of ECG signals. EMD is also used to extract features from ECG signals during cardiac auscultation using morphological signal processing [8]. An extension of EMD called multivariate empirical mode decomposition (MEMD) is proposed in [9] which showed the direct multi-channel signal processing using localization of frequency information. The decomposition is based on the assumptions: (1) the signal has at least two extrema: one maximum and one minimum; (2) the characteristic time scale is defined by the time lapse between the extrema; and (3) if the data were totally devoid of extrema but contained only inflection points, then it can be differentiated once or more times to reveal the extrema. Final results can be obtained by integration(s) of the components. The advantage of

EMD over the Fourier analysis is that, every IMF is considered as a band signal showing different temporal scale data [6].

The process of EMD is presented briefly for completeness [10, 11]. For a given signal, $x(t)$, local highest and lowest points are found as a first step of EMD. A cubic spline curve is used to connect all the local highest and lowest points (extrema) giving upper envelope $x_u(t)$ and low envelope $x_l(t)$. Further, to observe the values at every point of envelopes the mean value curve is calculated. The mean value, $m_1(t)$ is defined as follows for two envelopes

$$m_1(t) = \frac{x_u(t) + x_l(t)}{2} \quad (1)$$

In this way, the value for the first proto- IMF, $h_1(t)$ can be calculated as follows:

$$h_1(t) = x(t) - m_1(t) \quad (2)$$

The process of obtaining the IMF is generally known as shifting process. This process is used to cut off the riding waves and to ensure the symmetry of wave-profiles. It is a recurring process. For this, during the next shifting process $h_1(t)$ is considered as original data and the notation for second proto- IMF is as follows:

$$h_{11}(t) = h_1(t) - m_{11}(t) \quad (3)$$

The shifting process is repeatable for k times. It is continued till $h_{1k}(t)$ is considered as an IMF.

$$h_{1k}(t) = h_{1(k-1)}(t) - m_{1k}(t) \quad (4)$$

The relation between first IMF $c_1(t)$ and $h_{1k}(t)$ can be defined as follows:

$$c_1(t) = h_{1k}(t) \quad (5)$$

Furthermore, it is mandatory to use some stopping criteria to end the shifting process so that it can ensure the continuity of sufficient physical sense of both amplitude and frequency modulations of IMF elements. The size of standard deviation (SD) is the stopping criteria and can be designated as:

$$SD = \sum_{t=0}^T \frac{|h_{k-1}(t) - h_k(t)|^2}{h_{k-1}^2} \quad (6)$$

The first IMF $c_1(t)$ is achieved for the smaller value of SD compared to threshold value. The value of residue is computed using the following equation:

$$r_n(t) = x(t) - \sum_{i=1}^n c_i(t) \quad (7)$$

The following flow chart can briefly explain the whole method:

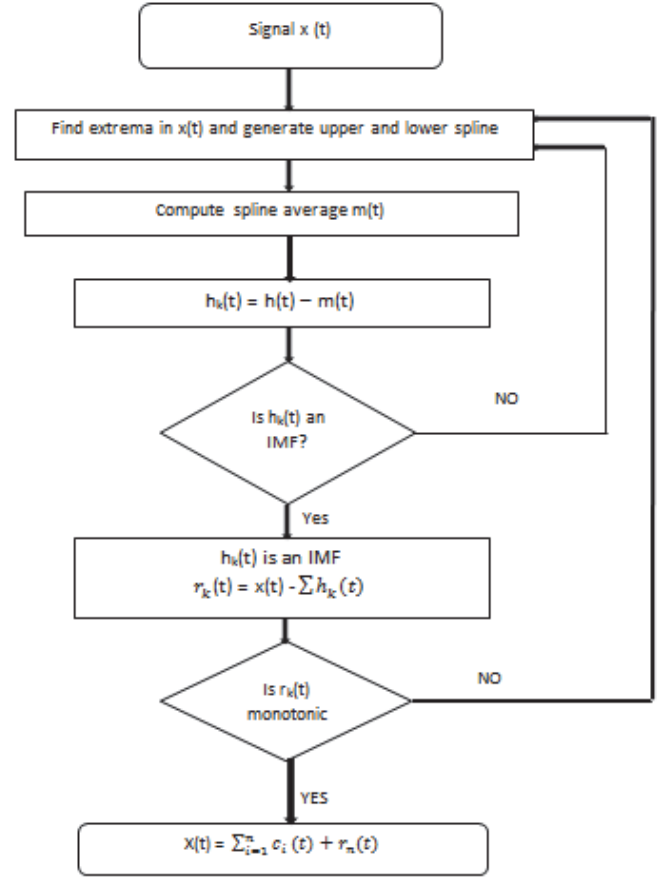


Figure 1: Flow chart of EMD algorithm [11]

III. METHODOLOGY

In this work, EEG signals were recorded from 64 Ag/AgCl scalp electrodes against a reference electrode fixed on the tip of the ear lobe. The electrodes were arranged following the international 10-20 system [12]. The subject was asked to perform three movements (left hand, right hand and both hands) after a series of stimuli. Each trial started with a short pause indicating that the trials are about to begin. Two seconds later a message appeared on a computer screen in front of the subject, indicating which movement was to be done. The movements asked for were chosen at random among the three movements studied. A total of 100 trials of each class of movement was recorded in blocks of 5 sessions. Here, we only use a subset of these data, comprising the movements of the left hand and right hand.

The r square topography of left hand vs right hand using BC12000 is shown in figure 2. From this figure channels with the largest r square value was selected for the empirical mode decomposition algorithm. For the left-hand movement C4, C6, Cp4 and Cp6, whereas for the right-hand movement C1, C3, CCp1 and Cp3 were selected.

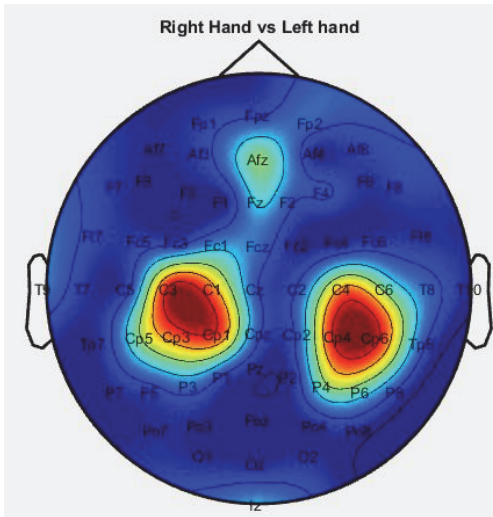


Figure 2: r square topography of left hand and right hand movement using BCI 2000 [13]

Signals from these channels are plotted in figure 3. Figure 4 shows the power spectrum of the channels selected. It is clear that high frequency noise and 60 Hz power line interference are present in the original EEG signals. Biomedical signals, especially EEG signals, are often contaminated with the 60 Hz line frequency interference from wires, light fluorescents and other equipment which are captured by the electrodes and acquisition system. The light of fluorescents usually causes artificial spikes in the EEG. They are distributed in several channels of EEG and can introduce error in the analysis of the record [14].

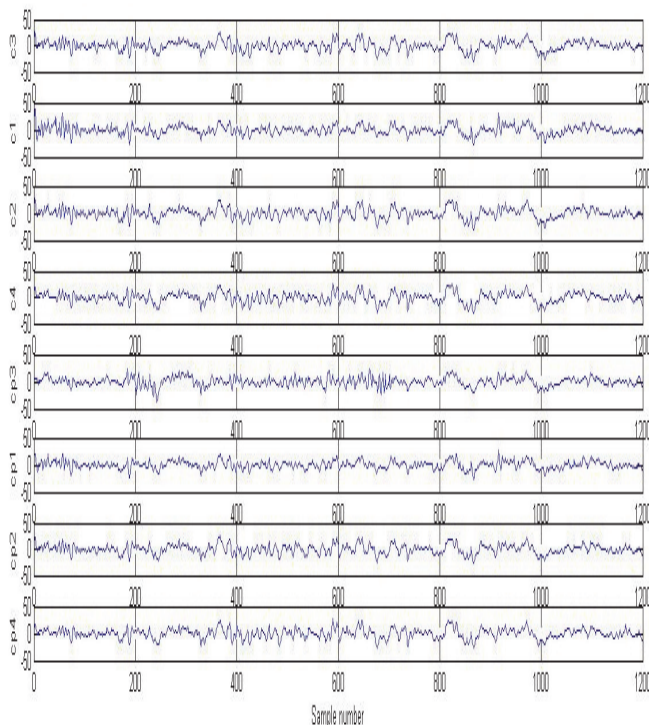


Figure 3: Original EEG signals from selected channels

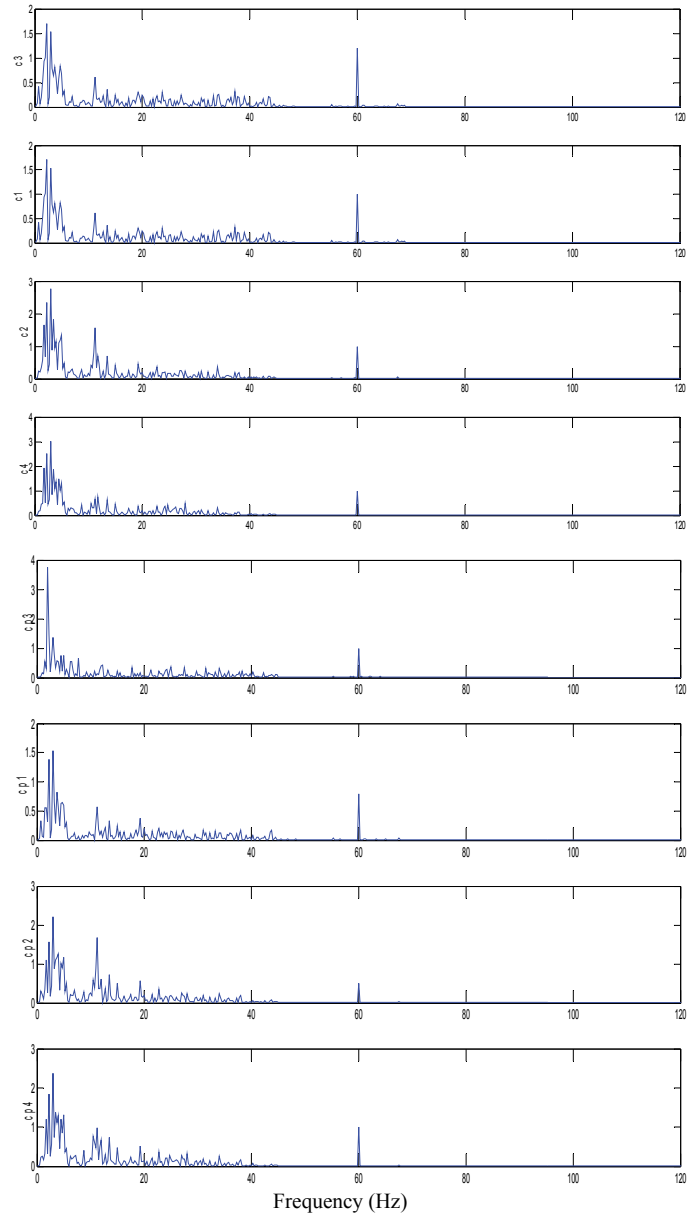


Figure 4: Power spectrum of original signals

IV. RESULTS AND DISCUSSION

The channel C3 EEG signal decomposed by the EMD is shown in figure 5. Signal from C3 channel is broken down in time domain. Here C3 electrode corresponds to the right-hand movement. The original signal is shown in the top after that the intrinsic mode functions are plotted with low to high order. Non-linear and non-stationary signals are analyzed using this algorithm. For the left-hand movement, the channel stimulated is Cp4 which is also decomposed using EMD in the similar way and presented in figure 6. Though original C3 and Cp4 signals look similar, the differences are manifested in the component IMFs.

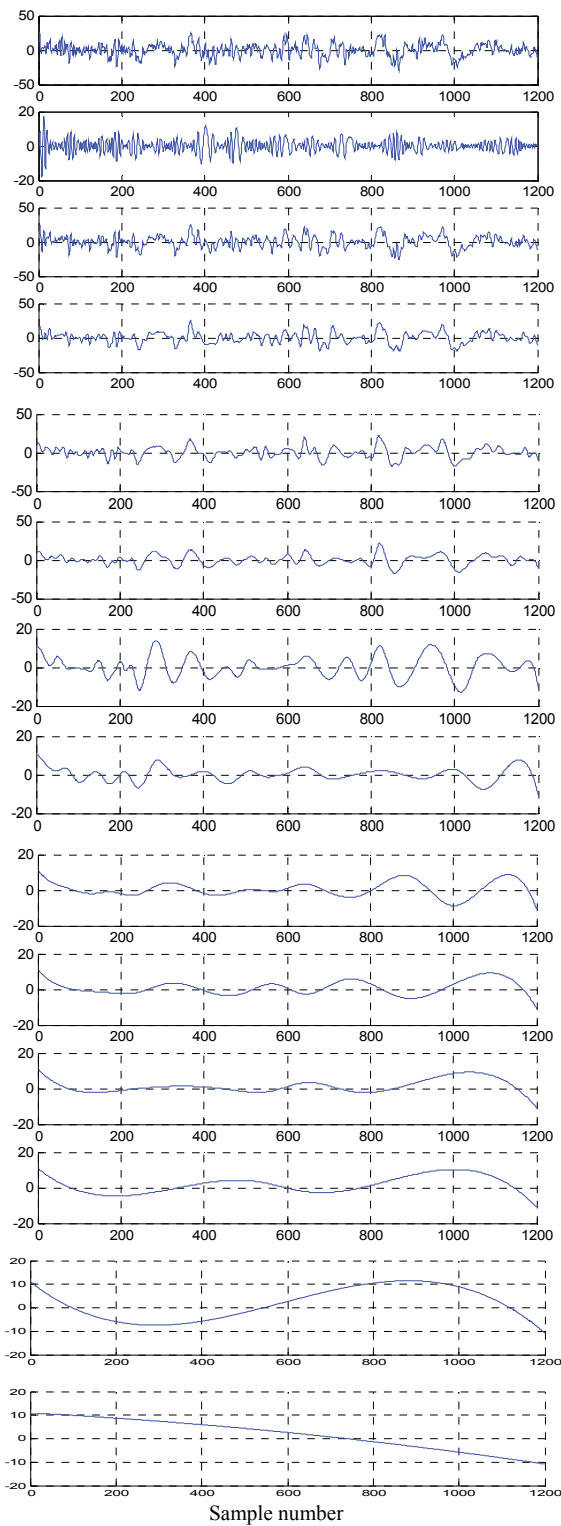


Figure 5: EMD of channel C3 signal (right hand movement) (First graph in the figure is the original signal and the last one is residue, remaining are the IMFs 1 to 12)

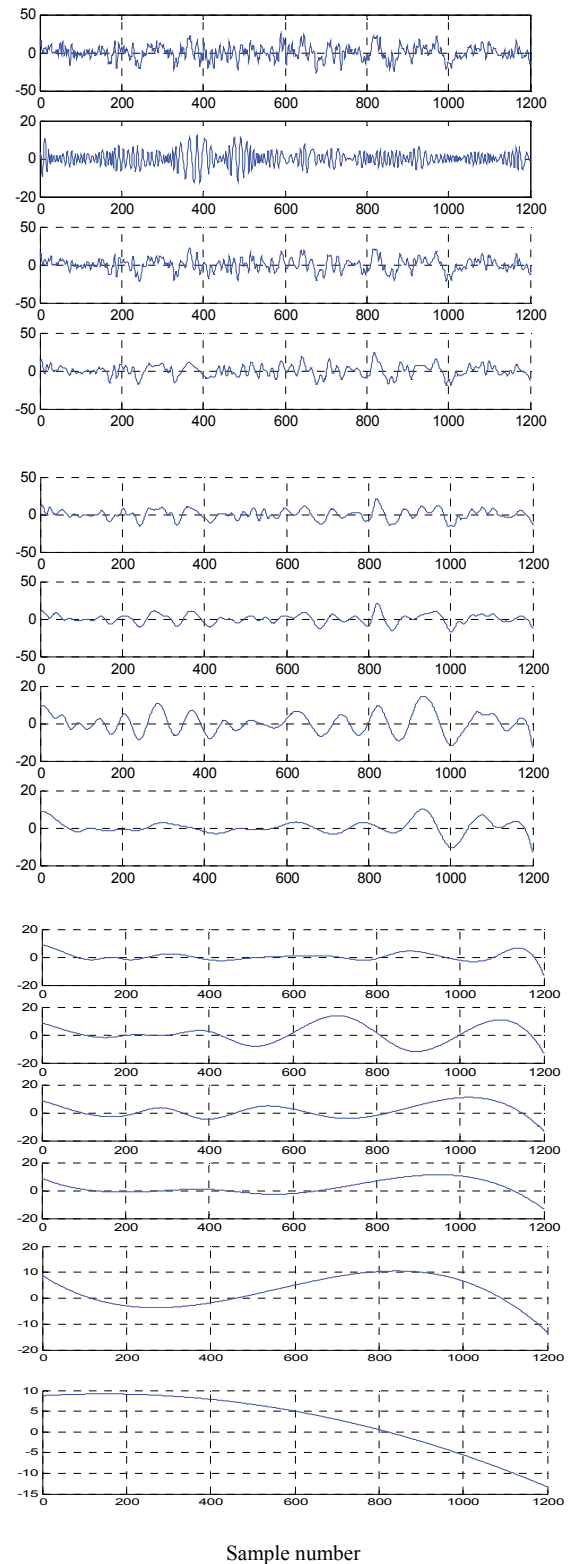


Figure 6: EMD of channel Cp4 signal (left hand movement) (First graph in the figure is the original signal and the last one is residue, remaining are the IMFs 1 to 12)

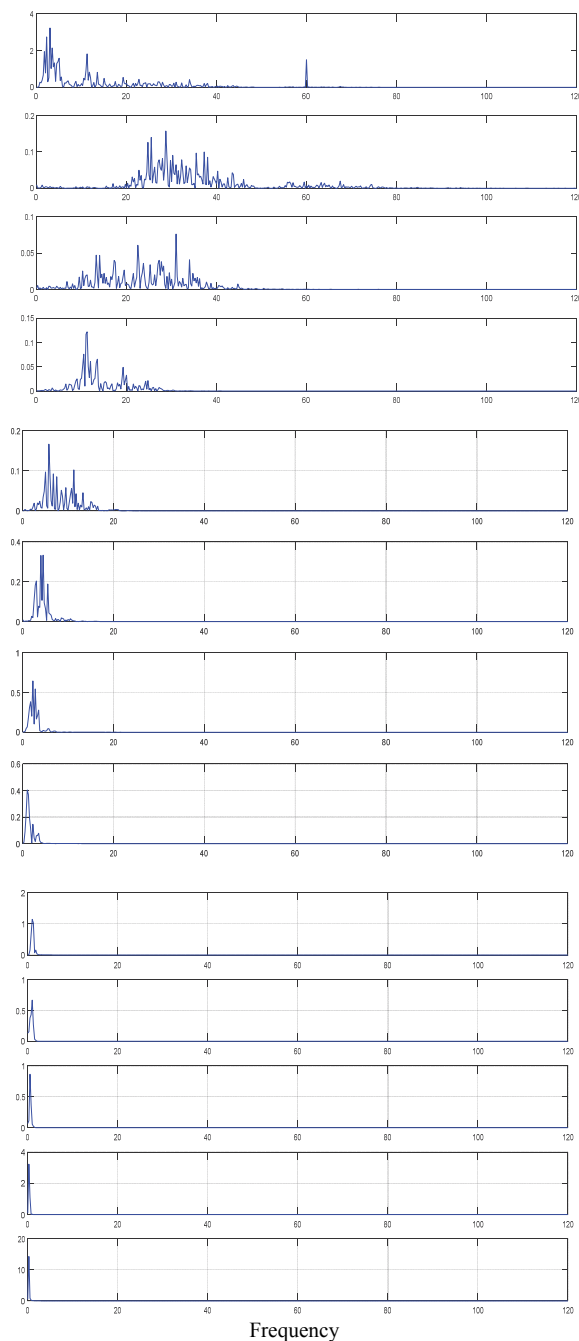


Figure 7: Power spectra of intrinsic mode function of channel C3

The power spectra of the EMD components of EEG signals from channels C3 and Cp4 are shown in figures 7 and 8 respectively. The 60 Hz power line frequency, present in original signals, is eliminated in the spectra of EMD components. The differences between C3 (Right hand movement imagery) and Cp4 (left-hand movement imagery) are evident in both time domain EMD components and their corresponding frequency spectra, as shown in figures 6 and 7, and figures 7 and 8 respectively.

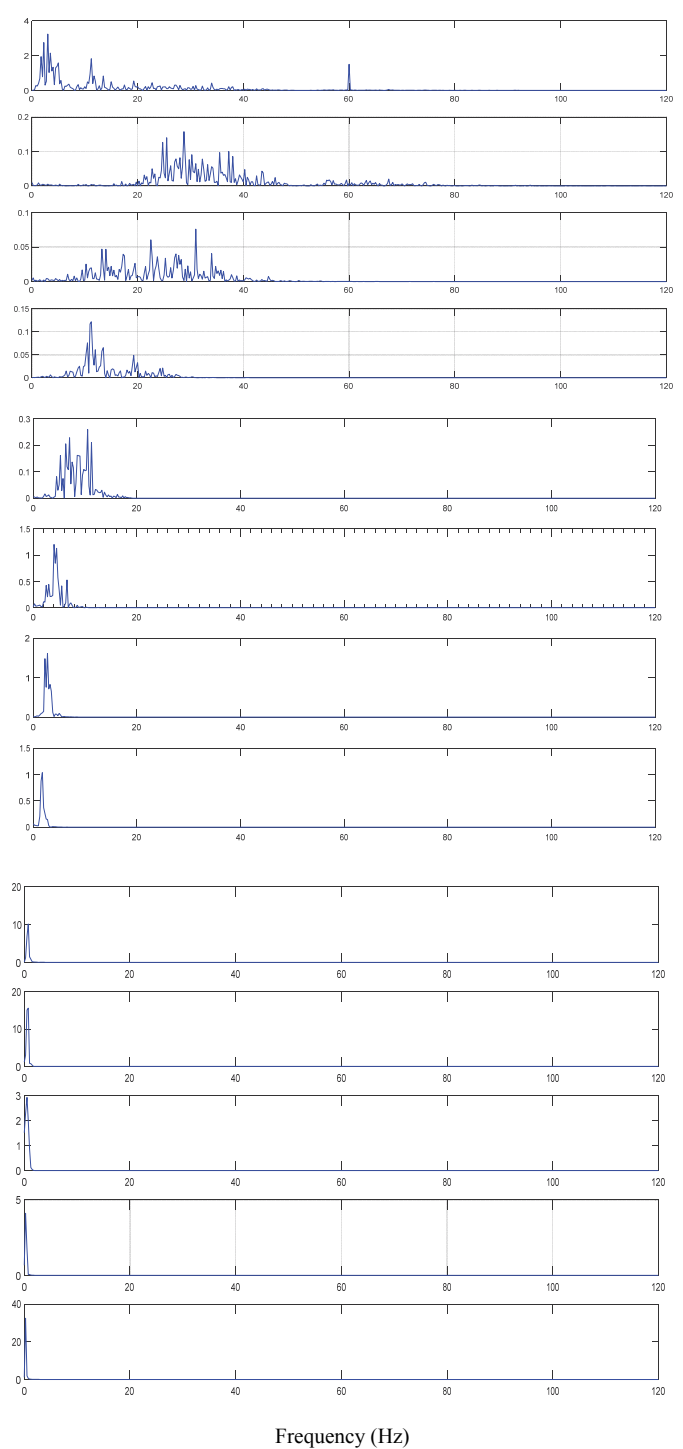


Figure 8: Power spectra of intrinsic mode function of channel Cp4

The power spectra of the IMFs cover physiologically meaningful frequency ranges corresponding to different rhythms. For example, the power spectra of IMFs 3-7 (rows 4-8 of figures 7 and 8) cover the rhythms β , μ , α , θ , and δ with some overlaps. The time domain IMFs can be processed further to extract features to distinguish different MI activities.

V. CONCLUSION

The study presents initial results of processing actual EEG signals using empirical mode decomposition (EMD) for potential applications in motor imagery based brain-computer interface (MI-BCI) systems. The EEG signals, traditionally characterized as noisy, nonlinear and non-stationary, can be processed using EMD to get signal components (intrinsic mode functions- IMFs) in the physiologically relevant frequency domains for MI-BCI applications. The decomposed signal components, IMFs, can be further processed to extract relevant features for classification and activation of MI activities for translating to actionable signals for remote operation of devices. Further work is in progress to develop an MI-BCI system using the proposed approach.

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